Quantitative Coefficient Survival, Supported Restriction, and Kernel-Retaining Bounds in Quillen Homology

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Abstract

We study the coefficient-sensitive steps of the manuscript *Rational homology and Quillen's conjecture*. For the building of an m-dimensional $\mathbb{F}_p$-space we prove an explicit sharp coefficient-norm opposite-chamber estimate with a boundary-defect term, and transfer it through the manuscript's fixed-complement restriction without merging coefficients. A supported exact correction yields a quantitative certificate only when its cost is smaller than the surviving coefficient. We also retain the two-dimensional relation label in the manuscript's marked-factor argument, obtaining a conditional positive kernel bonus in its graded dimension inequality. Actual subgroup examples distinguish nonzero restricted cycles from nonzero homology classes. Classical building, chain-complex and PBW tools are credited; the source's global finite-group classification and realization arguments remain separate inputs, not consequences of finite tests. Contribution assessment: C3/5, the report proves a source-specific defect budget for coefficient-preserving restriction and a kernel-retaining improvement of the marked-factor bound, without independently certifying the global conjecture.

1 Problem and scope

For a finite group G and a prime p, $\mathcal{A}_p(G)$ is the poset of nontrivial elementary abelian p-subgroups, ordered by inclusion. Write $m_p(G)$ for their maximum $\mathbb{F}_p$-rank and $O_p(G)$ for the largest normal p-subgroup. The source claims

$$O_p(G) = 1 \quad \Longrightarrow \quad \widetilde{H}_*(\mathcal{A}_p(G); \mathbb{Q}) \neq 0 \tag{1}$$

for every finite group and prime. This is the rational-homology form of Quillen's conjecture, not a statement about one selected coefficient matrix. The source uses group-theoretic reductions, decorated frame cycles, a maximal-index restriction, and propagation from a component. Here the global claim is attributed to that manuscript. We do not certify it independently.

All chain groups below are finite-dimensional augmented simplicial chain groups over $\mathbb{Q}$, with simplices oriented by their increasing poset order. Thus $\partial[x_0 < \cdots < x_q] = \sum_{i=0}^{q} (-1)^i [x_0 < \cdots < \widehat{x}_i < \cdots < x_q]$, $\partial[x] = [\varnothing]$, and $\widetilde{C}_{-1}(\varnothing) = \mathbb{Q}$. The coefficient norm is the maximum absolute coefficient in this specific oriented-simplex basis. It is not an invariant norm under arbitrary moment coordinates. A full rank-m flag in $\mathcal{A}_p(G)$ has vertices of ranks $1, \ldots, m$ and degree $m - 1$.

The main additional result is a quantitative version of the source's opposite-chamber detection and restriction. It treats exact cycles and inexact chains separately, gives a correction threshold,

and records which geometric support must survive correction. A secondary result strengthens one graded inequality by remembering the fixed relation plane. Neither changes the source's classification hypotheses, supplies missing faithful configurations, or improves its universal exponential dimension constant automatically.

2 Quantitative opposite-chamber detection

Let $V = \mathbb{F}_p^m$, $m \geq 2$, and let $\mathcal{B}(V)$ be its proper nonzero subspace poset. Fix $0 = F_0 < F_1 < \cdots < F_m = V$, $\dim F_j = j$. A chamber $W = (W_1 < \cdots < W_{m-1})$ is opposite to F if

$$\dim(W_i \cap F_j) = \max(0, i + j - m), \qquad 0 \leq i, j \leq m,$$

where $W_0 = 0, W_m = V$. Write $\mathrm{Opp}(F)$ for these chambers and $N = \binom{m}{2}$. For $k \geq 0$ put

$$b_k = \frac{p^k - 1}{p - 1}, \qquad \eta_k(A, \delta) = p^{-k} A - \frac{1 - p^{-k}}{p - 1}\delta. \tag{2}$$

These quantities use the actual field size p; there are $p + 1$ completions of every panel.

Theorem 2.1 (A coefficient and defect estimate)**.** *For every $z \in \widetilde{C}_{m-2}(\mathcal{B}(V); \mathbb{Q})$,*

$$\|z\|_\infty \leq p^N \left\|z|_{\mathrm{Opp}(F)}\right\|_\infty + b_N \|\partial z\|_\infty . \tag{3}$$

More precisely, if a chamber W has coefficient of absolute value A and relative permutation w, there is an opposite chamber with coefficient of absolute value at least $\eta_k(A, \delta)$, where $k = N - \mathrm{inv}(w)$ and $\delta = \|\partial z\|_\infty$. A negative lower bound is understood as vacuous. For exact cycles, restriction to opposite-chamber coefficients is injective and its inverse on its image has coefficient norm at most p^N.

Proof. Set $r_{ij} = \dim(W_i \cap F_j)$. The differences $r_{ij} - r_{i-1,j}$ are zero and then one as j increases, so their thresholds form a permutation w with $r_{ij} = |\{a \leq i : w(a) \leq j\}|$. If w is not decreasing, choose an ascent $w(i) = a < w(i + 1) = b$. In the two-dimensional quotient W_{i+1}/W_{i-1} the reference flag induces zero, one line for $a \leq j < b$, and the full quotient. The current completion is the unique one using that line. Each of the other p completions interchanges a, b and increases the inversion count by exactly one.

The coefficient of this panel in ∂z, up to its common orientation sign, is the sum of its $p + 1$ chamber coefficients. Consequently one of the other completions has coefficient a' with

$$|a_W| \leq p|a'| + \delta.$$

Repeat, always choosing a largest absolute coefficient among those p alternatives. The relative permutation changes by a specified ascent swap even if a selected coefficient is zero. After exactly k steps it is decreasing, hence opposite. Iteration gives $A \leq p^k |a_{\mathrm{final}}| + b_k \delta$. This proves the pointwise bound. Apply it to a largest coefficient, use $k \leq N$, and enlarge the nonnegative terms to obtain (3). Setting $\delta = 0$ proves the last assertion. □

For $m = 2$ the only panel is the empty simplex and the proof is just the augmentation equation. For $m = 1$ the building is empty; its single augmented coefficient is detected with constant one and no defect term. The source proves the qualitative exact-cycle assertion by an inversion maximum, following the classical apartment-basis framework of Solomon and Björner. The increment here is an explicit coefficient loss, a path-dependent refinement $k \leq N$, and the tolerated boundary error. The next classical weighted-chamber construction identifies the exact constant in this particular norm; it is not a new principle of building homology.

Proposition 2.2 (The coefficient-norm constant is sharp)**.** *For every $m \geq 2$ and prime p, the least possible coefficient of $\left\|z|_{\mathrm{Opp}(F)}\right\|_\infty$ in (3), even when restricted to exact cycles, is $p^{\binom{m}{2}}$.*

Proof. Assign to every chamber the rational coefficient $a_W = (-1/p)^{\mathrm{inv}(w_F(W))}$. Fix any panel, with its ranks other than i held fixed. Its two varying thresholds are two distinct numbers $a < b$. The induced flag on the two-dimensional quotient has one intermediate line, exactly as in Theorem 2.1. There is one completion with thresholds (a, b) and p completions with thresholds (b, a). All other thresholds are fixed; adjacent interchange changes the inversion count by one. Thus the panel sum is $(-1/p)^\ell + p(-1/p)^{\ell+1} = 0$. This includes the augmented panel for $m = 2$, so the chain is a cycle. The reference chamber has inversion zero and coefficient one. All coefficients have magnitude at most one, while precisely the opposite chambers have inversion N. Their magnitudes are p^{-N}. The ratio is therefore p^N, as asserted. □

Example 2.3 (Rank two sharpness)**.** There are $p + 1$ lines. Let the reference line have coefficient p and each of the other p lines coefficient -1. This is an augmented zero-cycle, the other lines are precisely the opposite chambers, and (3) is equality with constant p. For a defect example give the reference line coefficient $p + \delta$ and the others -1, with rational $\delta \geq 0$. Both terms of the estimate are then attained. Thus even this smallest building rules out a defect-free estimate for arbitrary chains.

3 Restriction without coefficient collisions

We give the support hypotheses in full, since neither a quotient map nor arbitrary intersection with a subgroup preserves coefficients. Let M be finite, $H \leq M$, and $\alpha \in \widetilde{C}_{m-1}(\mathcal{A}_p(M); \mathbb{Q})$ be supported on full rank-m flags. It need not initially be a cycle. Let $\mathcal{E}(\alpha)$ be its supported top groups. Suppose

$$0 \leq c < m, \qquad [E' : E' \cap H] \leq p^c \quad (E' \in \mathcal{E}(\alpha)), \qquad [E : E \cap H] = p^c \tag{4}$$

for a specified supported E. Write $K = E \cap H$.

Theorem 3.1 (Quantitative fixed-complement restriction)**.** *Suppose first $0 < c < m$, choose a reference flag in E through K, and choose a supported chamber below E with coefficient magnitude A. Put $\delta = \|\partial\alpha\|_\infty$ and $k = \binom{m}{2} - \mathrm{inv}(w)$ for this chamber and reference flag. If $\eta_k(A, \delta) > 0$, one can construct a chain $\beta \in \widetilde{C}_{m-c-1}(\mathcal{A}_p(H); \mathbb{Q})$ such that:*

1. *its support consists of full flags of rank $m - c$;*
2. *a flag ending at K has coefficient magnitude at least $\eta_k(A, \delta)$;*
3. *$\|\partial\beta\|_\infty \leq \delta$.*

If $\partial\alpha = 0$, this is a nonzero cycle. It is a nonzero homology class if additionally $m - c = m_p(H)$. For $c = 0$ use $\beta = \alpha$ inside $\mathcal{A}_p(H)$, with no coefficient loss.

Proof. Collect all terms ending at E and delete that last vertex, giving $z_E \in \widetilde{C}_{m-2}(\mathcal{B}(E))$. Every boundary face retaining E receives contributions only from such terms. Hence $\|\partial z_E\|_\infty \leq \delta$, also in the augmented rank-two case. Theorem 2.1 produces an opposite chamber with positive coefficient magnitude at least $\eta_k(A, \delta)$. Its rank-c vertex R complements K, so $E = R \times K$. Fix its saturated prefix $\sigma = [R_1 < \cdots < R_c = R]$, $\mathrm{rank}\, R_i = i$.

Let R_σ retain simplices beginning with this exact prefix and delete its c vertices. On augmented chains,

$$R_\sigma \partial = (-1)^c \partial R_\sigma. \tag{5}$$

To check this, deleting a suffix vertex has the indicated sign shift. A simplex not beginning with σ cannot acquire that prefix by deleting a vertex: saturation leaves no rank available below R_1 or between consecutive prefix vertices. Each output suffix has a unique input with that prefix. Thus R_σ is coefficient extraction and is a contraction in the coefficient maximum norm.

Every continuation vertex B lies in a supported E', contains R, and satisfies

$$p^c = |R| \le [B : B \cap H] \le [E' : E' \cap H] \le p^c.$$

All these are vector-space indices inside E'. Equality implies $B = R \times (B \cap H)$. Therefore intersection with H on the generated support subcomplex is an injective simplicial map with inverse $T \mapsto RT$ on its image. This is the same fixed R for every vertex, not a collection of independently chosen complements. Set $\beta = (B \mapsto B \cap H)_* R_\sigma \alpha$. There are no coefficient collisions, including on boundary faces. Equation (5) proves the defect bound; the selected coefficient survives and ranks drop by c. In top degree of $\mathcal{A}_p(H)$ there are no chains one degree higher, so an exact nonzero cycle cannot be a boundary. □

The positivity threshold is $A > b_k\delta$. In particular the uniform sufficient condition $A > b_N\delta$ may be conservative. One should retain the actual inversion distance and actual initial coefficient when they are known. The maximum-index hypothesis in (4) concerns *all* supported top groups, including small nonzero coefficients. Dropping those coefficients before verifying the hypothesis changes the input chain and its defect.

4 Exact correction and the homology gate

A small boundary is not a cycle. The following finite statement makes the missing step explicit and keeps correction in the permitted support.

Proposition 4.1 (Supported correction with a retained witness)**.** *Let Y be the span of a specified collection of actual oriented q-simplices of $\mathcal{A}_p(H)$ and let $d = \partial|_Y : Y \to \widetilde{C}_{q-1}(\mathcal{A}_p(H); \mathbb{Q})$ include* all *boundary faces. Suppose a rational linear map $T : \operatorname{im} d \to Y$ and a constant $\kappa \ge 0$ satisfy $dT = I$ and $\|Tr\|_\infty \le \kappa \|r\|_\infty$. For $\beta \in Y$ with $\|d\beta\|_\infty \le \delta$,*

$$\widehat{\beta} = \beta - Td\beta, \qquad d\widehat{\beta} = 0.$$

If a specified coefficient of β has magnitude $a > \kappa\delta$, then $\widehat{\beta} \ne 0$ and that coefficient has magnitude at least $a - \kappa\delta$. Applied to Theorem 3.1, a sufficient condition is

$$A > \big(b_k + p^k\kappa\big)\delta. \tag{6}$$

Proof. The right-inverse identity kills the boundary exactly. The norm bound controls every changed coefficient. Substituting $a \ge \eta_k(A, \delta)$ gives (6). □

Such a right inverse exists algebraically for every finite d on its image, but a useful small κ is additional information. One may choose independent rational columns, solve on their image, and certify the resulting linear identities exactly. A numerical residual alone does not establish them. Nor may the boundary matrix omit faces that leave a preferred subsystem.

For a nonzero class in the *ambient* complex one still needs one of the following: $q = m_p(H) - 1$; a rational cochain λ annihilating every ambient boundary with $\lambda(\widehat{\beta}) \ne 0$; or a separate propagation theorem whose full support and group-theoretic hypotheses are met. Merely checking nonboundary status inside Y can be misleading, since Y need not contain all possible fillings.

If a correction is to feed the source's faithful-configuration propagation, Y must preserve the allowed full flags and the particular terminal faithful subgroup. An arbitrary least-squares correction is not enough. In changed moment coordinates $m = Ec$, both the boundary and the witness transform: $d^{\mathrm{new}} = E_{q-1} d E_q^{-1}$ and $\lambda^{\mathrm{new}} = \lambda E_q^{-1}$. The coefficient estimates above are estimates on c, not on unscaled m. This is a direct application of the original TR250 compatibility principle, not a new topological theorem from a finite invertible matrix.

5 Actual subgroup examples and negative controls

Let $p = 2$, $M = S_3 \times S_3$, and choose distinct transpositions a_1, a_2 in the first factor and b_1, b_2, b_0 in the second. We identify them with the corresponding elements of the direct product. Put $A_i = \langle a_i \rangle$, $B_j = \langle b_j \rangle$ and $E_{ij} = A_i B_j$. The source's product shuffle of two augmented zero-cycles is the explicit one-cycle

$$\alpha = \sum_{i,j=1}^{2} (-1)^{i+j} \big([A_i < E_{ij}] - [B_j < E_{ij}] \big). \tag{7}$$

Each summand has boundary $[B_j] - [A_i]$, and summing first over i or j proves $\partial\alpha = 0$. All its eight flags are actual, distinct subgroup flags.

Take $H = S_3 \times 1$. Every E_{ij} has intersection A_i and index two in H. Residue at the saturated vertex B_1 followed by intersection gives, up to one overall sign, $\beta = [A_1] - [A_2]$. The poset $\mathcal{A}_2(H)$ is three isolated points, $m_2(H) = 1$, and β represents a nonzero reduced zero-homology class. Here the restriction theorem applies with $m = 2, c = 1$ and its top-degree conclusion is legitimate.

Now instead take $H' = S_3 \times B_0$. Since b_0 is distinct from b_1, b_2, the same intersection indices and the same restricted chain occur. However $m_2(H') = 2$, and this nonzero zero-cycle bounds. Indeed, setting $F_i = A_i B_0$, the actual one-chain

$$\gamma = -[A_1 < F_1] + [B_0 < F_1] + [A_2 < F_2] - [B_0 < F_2] \tag{8}$$

has $\partial\gamma = [A_1] - [A_2]$. The normal central subgroup B_0 also gives the poset contraction $D \le DB_0 \ge B_0$. Thus neither a nonzero coefficient nor exact boundary cancellation by itself proves the ambient homology claim.

These small groups make the support issue concrete. In $S_3 \times S_3$ there are 15 rank-one and 9 rank-two elementary subgroups, with 27 incidences and no higher ranks. The incidence graph is connected, so its first rational Betti number is $27 - 24 + 1 = 4$. For $S_3 \times B_0$ the corresponding graph has 10 vertices and 9 edges and is a tree. These counts can be verified by multiplying the 36 actual permutations, not by postulating an abstract matrix with the desired rank.

There is also a simple supported-repair example. On the three-point complex of S_3, let $\beta_\varepsilon = (1 + \varepsilon)[A_1] - [A_2]$. Its augmented boundary is ε. The right inverse $T(r) = r[A_0]$, where A_0 is the third point, has $\kappa = 1$. Thus $\widehat{\beta}_\varepsilon = (1 + \varepsilon)[A_1] - [A_2] - \varepsilon[A_0]$ is exactly closed, and its coefficient at A_2 is still -1. A coordinate-specific bound can therefore outperform the global norm bound: if the correction row at the witness is zero, that witness incurs no loss even when $\kappa > 0$.

6 Retaining the relation plane in the frame estimate

This section is conditional on the source's actual good-support graded construction, not on an arbitrary ungraded presentation. We describe the input precisely. A type $[W]$ is an isomorphism

or isometry type of an admissible finite-field space. Multiplication is direct sum; in form cases it is orthogonal direct sum. For an equivariant family of finite vector spaces $M(W)$ write

$$\mathsf{M} = \sum_{[W]} \frac{\dim M(W)}{|G_W|}[W], \qquad D[W] = (\dim W)[W].$$

Here G_W is the full linear or isometry group specified by the source. Ordered decompositions have stabilizer $G_A \times G_B$, so multiplication of these normalized series counts $\bigoplus_{W=A\oplus B} M(A) \otimes N(B)$. Inequalities below are coefficientwise in this type basis, not just after summing the two discriminant types.

The source starts with odd line letters and even plane relations, forms a Lie superalgebra P and its enveloping algebra U, and uses an actual relation-label module R, with each label supported on an admissible plane of dimension two. Put $C = \ker(U \otimes R \to U \otimes V)$. Its established structural inputs are

$$\mathsf{U} = \exp(\mathsf{P}), \qquad \mathsf{U}(1 - v + w) = 1 + \mathsf{C}, \qquad \operatorname{gr} C \subset \operatorname{Sym}_{\text{super}}(P) \otimes R. \tag{9}$$

The last inclusion uses the induced PBW filtration on C. Every good monomial has pairwise independent factor supports and a relation plane independent of them. Bad overlapping supports are discarded by the source grading; this restriction is indispensable.

Proposition 6.1 (A two-dimensional kernel bonus)**.** *Under these structural inputs and the source's multiplicative leading module-monomial order,*

$$(D-2)\mathsf{C} \;\geq\; (D\mathsf{P})\mathsf{C}. \tag{10}$$

Writing $J = -\log(1 - v + w)$, if DJ is coefficientwise nonnegative,

$$D\mathsf{P} \geq (1 + \mathsf{C})DJ + 2\mathsf{C}. \tag{11}$$

In a type W of dimension $h > 0$, this says

$$\frac{\dim P(W)}{|G_W|} \geq J_W + \frac{2\mathsf{C}_W + (\mathsf{C}DJ)_W}{h}. \tag{12}$$

Proof. Fix W and bases of all $P(A)$ for admissible $A \leq W$. In each good component of $\operatorname{gr} C$, row reduction gives a set of distinct leading module monomials whose cardinality is its dimension. If $W = A \oplus B$, a basis vector $p \in P(A)$ times a leading monomial m in degree B is a nonzero leading monomial in degree W. Indeed independent supports forbid a repeated odd factor, prevent any exterior product from vanishing, and preserve distinct terms. The associated graded space is a submodule, and the monomial order is multiplicative. Mark this appended p with weight $\dim A$. The marked output recovers p, m, A, B, including the relation label.

For a fixed unmarked output, all markable factors occupy direct summands in the complement of its fixed relation plane. Their total weight is therefore at most $h - 2$, not merely h. Counting the marked outputs gives

$$\sum_{W=A\oplus B} (\dim A) \dim P(A) \dim C(B) \leq (h-2) \dim C(W).$$

After normalization this is (10). Components of dimension below two vanish because there is no relation label there. From (9), formal differentiation gives the exact identity

$$D\mathsf{P} = (1 + \mathsf{C})DJ + D\mathsf{C} - \mathsf{C}D\mathsf{P}.$$

Use (10) without dividing by any signed series. This proves (11), and taking the type coefficient gives (12). □

The source's marked-factor proof already observes that the relation plane is present, but then upper-bounds the total mark weight by h. The refinement retains its exact reserved dimension. Its published bound remains valid; this is a sharpening, not a correction to exactness or PBW. We do not assert $C(W) \neq 0$ in every type. If the kernel vanishes in the relevant lower degrees, the extra terms vanish and the original logarithmic bound is unchanged.

The proof has a limited extension: when a compatible relation module has labels of one fixed positive dimension d and the same good-support and PBW premises are proved, replace 2 by d throughout. For mixed label dimensions at least d_0, the coarser bonus $d_0\mathsf{C}$ follows. Neither statement supplies the PBW structure for a new presentation. In particular an ordinary exterior algebra without support separation cannot use the nonvanishing-multiplication argument.

For a certified kernel coefficient $\mathsf{C}_W = c$ at dimension h, the direct improvement of the normalized Lie coefficient is at least $2c/h$, with a further convolution contribution from smaller types. This is an exact conditional numerical rule. Computing c still requires the actual relation map and exact kernel rank; a model monomial module is only a check of the counting argument, not a calculation of that source kernel.

7 How the local results meet the source proof

The manuscript's architecture is more restrictive than a sequence of dimension estimates. Each stage has an actual-chain or group condition which a moment representation must retain.

7.1 Frames, decoration counts and actual coefficients

The source's sign group on an h-line frame is $\mathbb{F}_p^h/\langle(1,\dots,1)\rangle$, of rank $h-1$. Its local coefficient space has dimension $(h-1)!$. Plane relations cancel merger faces. The free Lie superalgebra description keeps the sign from odd line letters, and the good-support grading discards overlapping supports. Replacing this by an ordinary commutative polynomial ring would change the identities used in Proposition 6.1.

For ordinary decorations with completion counts k_i, the source starts with at least Nq_hF coefficient parameters, $q_h = (h-1)!$, and imposes at most Nq_h/k_i independent conditions of type i. Its dimension lower bound is $Nq_h(F - \sum_i k_i^{-1})$. This remains a dimension of *actual cycles* only because the source separately proves injectivity of the map from data to tested full-flag coefficients. A positive formal parameter count is not a substitute for that injectivity.

The quaternion decoration has a two-dimensional local space $\{(c_\ell)_{\ell\le T} : \sum_\ell c_\ell = 0\}$ on the three lines of $T \cong C_2^2$. Hence its relative loss is $3/(2k_T)$, not $3/k_T$. With an optional field axis it gives $2Nq_h(F - 3/(2k_T) - 1/k_B)$. This valid source factor must not be lost in a scalar coefficient compression. The present restriction estimate acts *after* such a cycle and an actual supported top subgroup have been found. It neither constructs the compatible data nor supplies their counts.

7.2 Group-specific dependencies

Source module	Retained mathematical obligation
Reductions	Minimal-order counterexample, the classical component list at two, and the unitary-extension dimension property at odd primes.
Propagation	Faithful LA, $O_p(C_G(LA)) = 1$, centralizer and strict-overgroup exclusions, and a saturated supported coefficient.
Frames	Plane relations, actual subgroup faces, super PBW, type-wise normalization and positivity of DJ.
Chain procedures	Split-product signs, injective coefficient detection, maximum index on every supported top group.
Linear/unitary	Projective commutator pairing, scalar normalization, semilinear descent, compatible frames and determinant-coset counts.
Symplectic	Quaternion lifts commute exactly with the normalized field lift; the multiplicity form is symmetric. Cycles already lie in H.
Orthogonal	Determinant square class, spinor norm, triality naturality, allowed colors and maximal-index restriction.
Exceptional	Three branches for $\mathrm{PSL}_4(5)$, a direct Euler congruence, and the imported equivalent-poset theorem.

In the linear and unitary construction at two, every supported intersection $A' = E \cap H$ must generate the same H over L and belong to the maximized faithful family. In the orthogonal small-field case, the selected admissible frame attains the *maximum* index while other supported frames may have smaller indices. Theorem 3.1 preserves this useful generality; replacing maximum index by an assumed constant index would unnecessarily narrow that application.

At odd primes the source additionally proves the elementary rank needed to make its cycle top-dimensional. At two its faithful-family rank need not equal $m_2(G)$, and propagation rather than top degree is essential. The example (8) shows exactly why these two conclusions cannot be interchanged.

7.3 Propagation is an additional theorem, not a repair rule

For clarity, the source propagation mechanism takes a simple component L and an elementary $A \leq N_G(L)$ with $A \cap L \neq 1$, LA faithful on L, and $O_p(C_G(LA)) = 1$. It also requires that the elements of order p in $C_L(A)$ belong to A, and excludes strict elementary overgroups $D > A$ that remain faithful with $O_p(C_G(LD)) = 1$. A cycle on $\mathcal{A}_p(LA)$ must have a nonzero saturated coefficient ending at this very A.

The source deletes certain overgroups using contractible upper intervals, combines the cycle with an inductively nonbounding centralizer cycle by the signed product shuffle, and takes a residue. A centralizer retraction with left inverse then prevents the product from bounding. The algebra of the residue is local, but the deletion, faithfulness, centralizer core and induction hypotheses are not consequences of a small residual. They remain imported obligations when a repaired chain is used as input.

The exceptional $\mathrm{PSL}_4(5)$ branch additionally uses an equivalent poset with crossed comparisons. A no-comparison condition and an augmented Mayer–Vietoris argument inject the homology of a join. None of the present finite correction statements replaces that equivalent-poset theorem or the group-specific verification.

8 Original Ram proofs and the role of coordinates

The original TR250 source snapshot was freshly read in its physical space and compatible-coordinate section, lines 116–280, and its topology-transfer section and adjoining example, lines 724–887. The coordinate-transfer theorem explicitly preserves kernels, images and cohomology under invertible maps, while the topology theorem additionally requires

$$d_f P = P d_c, \quad d_c R = R d_f, \quad RP = I, \quad I - PR = d_f H + H d_f.$$

On a closed cochain x, the last identity makes $x - PRx$ exact. Commutation alone does not do so. Its triangular-ring example changes the first cohomology from one to zero when a face is filled, even though both complexes satisfy $d^2 = 0$. We use the same logical distinction here with homological degrees reversed: coefficient preservation and boundary cancellation do not prohibit an ambient filling.

The original also separates the physical mass matrix from incidence and orientation. For the present coefficient estimates, the oriented-simplex maximum norm is specified directly. If a Ram moment bank is used to compress a finite collection of actual subgroup chains, its reconstruction norm must be included; it cannot be discarded because an exact inverse exists. Proposition 4.1 is a finite admissible correction with a stated norm cost. It does not use point evaluations on an unspecified infinite space or assert uniform conditioning over all finite groups.

The later Core Pattern 50 text, freshly read at lines 232–435, supplies a secondary comparison: a separating period bank must annihilate boundaries and be injective on the intended quotient; minimum correction requires feasibility in the allowed subspace. These classical facts are not counted as new M610 theorems. TR250's revised manuscript was located and hashed but not proof-read for this report; the original snapshot, not that uninspected revision, is the original-proof evidence. No claim of complete Ram-corpus reading is made.

9 Current literature and status

Primary records were refreshed on October 9, 2026. Díaz Ramos's *Quillen's conjecture and unitary groups*, arXiv:2303.15613v6, was revised July 22, 2026; its abstract states top-dimensional nonvanishing for the required unitary extensions with specified exceptions and the consequent odd-prime result. Piterman's July 29, 2026 preprint arXiv:2607.27500 treats components in characteristic p under inductive assumptions and describes the prime-two conjecture as open at that time [5, 6].

The EMS record for Piterman–Smith identifies the equivalent-poset article as *Journal of Combinatorial Algebra* 9 (2025), 265–387, with online publication August 26, 2024. Its abstract describes the extension of the reduction to the remaining odd primes and partial progress at two [7]. The source manuscript dated September 24, 2026 makes the stronger all-prime claim. Its repository README and release listing were checked as publisher metadata, not as independent peer review or formal verification of that claim [1].

Piterman–Welker's frame-complex paper provides adjacent geometric context; its abstract includes characteristic-zero vanishing under a field-size/dimension hypothesis, not the unrestricted all-group claim [8]. These external access scopes are abstracts, version records and publication metadata. Their full proofs were not independently audited here. The local panel proofs above are self-contained; the classical apartment-basis attribution to Solomon and Björner is retained from the source bibliography [3, 4]. No priority assertion for the explicit quantitative corollaries follows from this targeted search.

10 Evidence, uses and limitations

The fresh literal source reading covers all 15 cached text files, 5,033 lines, including all ten scientific sections (4,568 lines), the fixed-complement figure, main file, bibliography, glyph map and README. Each endpoint was hashed before its bounded numbered displays and compared after reading. The ledger records actual display intervals, merges overlaps without double counting, and gives unread complements. Reading every source line does not independently prove its deep external reductions, global classification, or all group-realization assertions.

The independent executable checks use exact integers, rational numbers and rational matrix ranks. They contain 58 fixtures and 8,896 assertions, with zero floating-point diagnostics:

- nine actual buildings: $m = 2$ for $p = 2, 3, 5, 7, 11$; $m = 3$ for $p = 2, 3, 5$; and $m = 4, p = 2$;
- 27 perturbed-chain fixtures tracing the ascent procedure from every chamber in those buildings;
- three actual permutation-group subgroup graphs, three shuffle, restriction or filling fixtures, and three exact supported corrections;
- twelve disjoint-support marking models and one symbolic differentiated Euler identity.

An assertion is one explicitly checked identity or inequality; a fixture is one declared parameterized experiment, not one matrix entry. The marking models check only the combinatorial weight budget, not the actual source kernel. No external or parent-agent tests are included in these totals.

The improvement is useful when a candidate chain comes from a large but finite exact or certified computation and a particular supported coefficient must survive restriction. The norm constant is optimal for the declared basis, so replacing it by a dimension-free constant would require additional structure. The path-dependent k can be much smaller than $\binom{m}{2}$; exact arithmetic or a zero witness-row correction can improve the practical budget further.

Costs are substantial. Enumerating all subgroups and flags can be prohibitive. The universal coefficient loss grows as $p^{m(m-1)/2}$, and an ill-conditioned supported correction may consume the whole witness. No runtime saving is claimed. The kernel bonus is explicit but may be zero, and computing actual type-wise kernels may cost more than using the source's uniform lower bound. Finite rank cannot certify that a proposed group configuration exists or that all exterior automorphisms have been handled.

Promising next steps are narrower than a new proof of the global conjecture. One can choose reference flags through K to reduce the inversion distance of a known large coefficient; certify a correction which fixes that witness row; or compute a few actual relation kernels with support and isometry type retained. Weighted panel norms may give better budgets for structured chains, but they need their own sharp reconstruction theorem. Any extension to buildings with unequal panel sizes must replace the fixed p-branch estimate by the actual panel multiplicities and track their products.

11 Conclusion

The source's qualitative supported-complement argument admits a sharp coefficient-norm estimate and an explicit boundary-error budget. With maximum index on the entire support, the fixed complement preserves the selected coefficient and introduces no additional coefficient collision. Exact supported repair is possible only with a verified cost, and a separate ambient nonboundary condition remains mandatory. Remembering the relation plane also strengthens the graded mark count by $2\mathsf{C}$ when the source's good-support PBW premises hold.

These are notable local refinements, not a replacement for the source's finite-group proof. The

existing source conclusions and constants used here are retained as valid inputs where stated; no source correction was required by this scoped analysis. The distinction between a nonzero chain, a cycle, a nonbounding cycle and a globally realizable faithful configuration is the central constraint throughout.

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